DSM A2 Part 2 Final Report: MongoDB + Neo4j Querying

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1 MongoDB Querying

Q1: Cohort Analysis of User Reviewing Behaviour

Cohorts are defined by the year in each user's yelping-since profile field. For each cohort, I computed mean and standard deviation of stars, mean review character length, mean useful votes per review, and star-bucket proportions.

Q1 Metric	**Value**
Highest mean-star cohort	2005 (3.903)
Highest mean useful-votes cohort	2004 (2.276)
Total cohorts	19

Table 1: Q1 summary statistics.

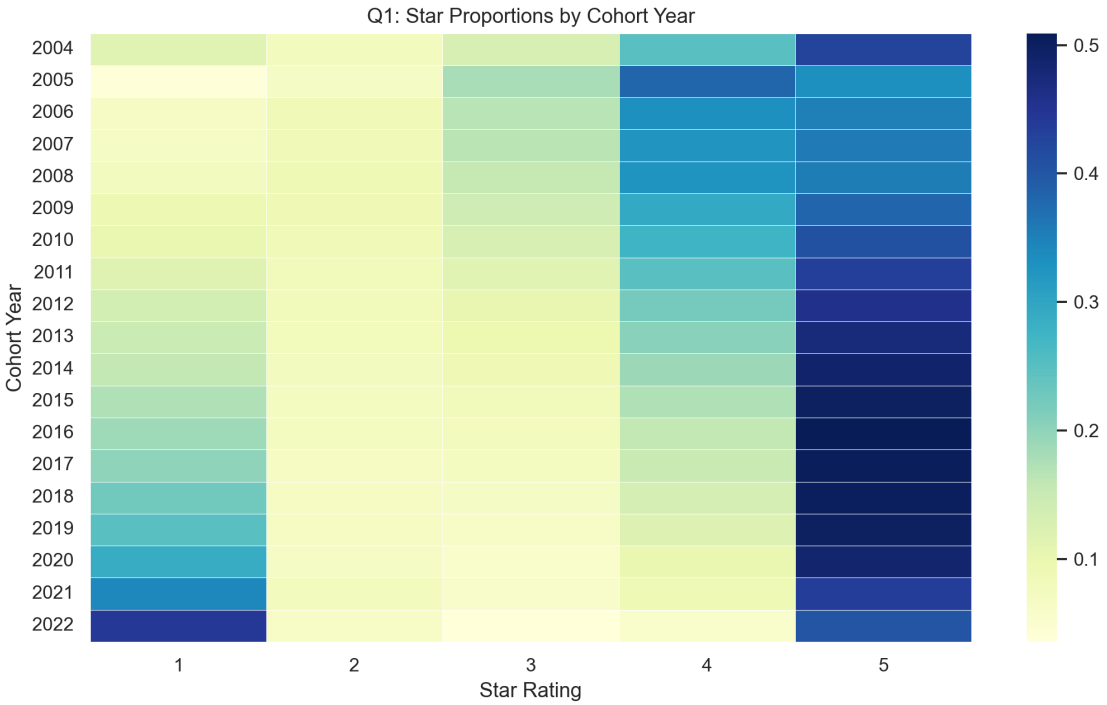

Figure 1: Q1 heatmap: star proportions by cohort year.

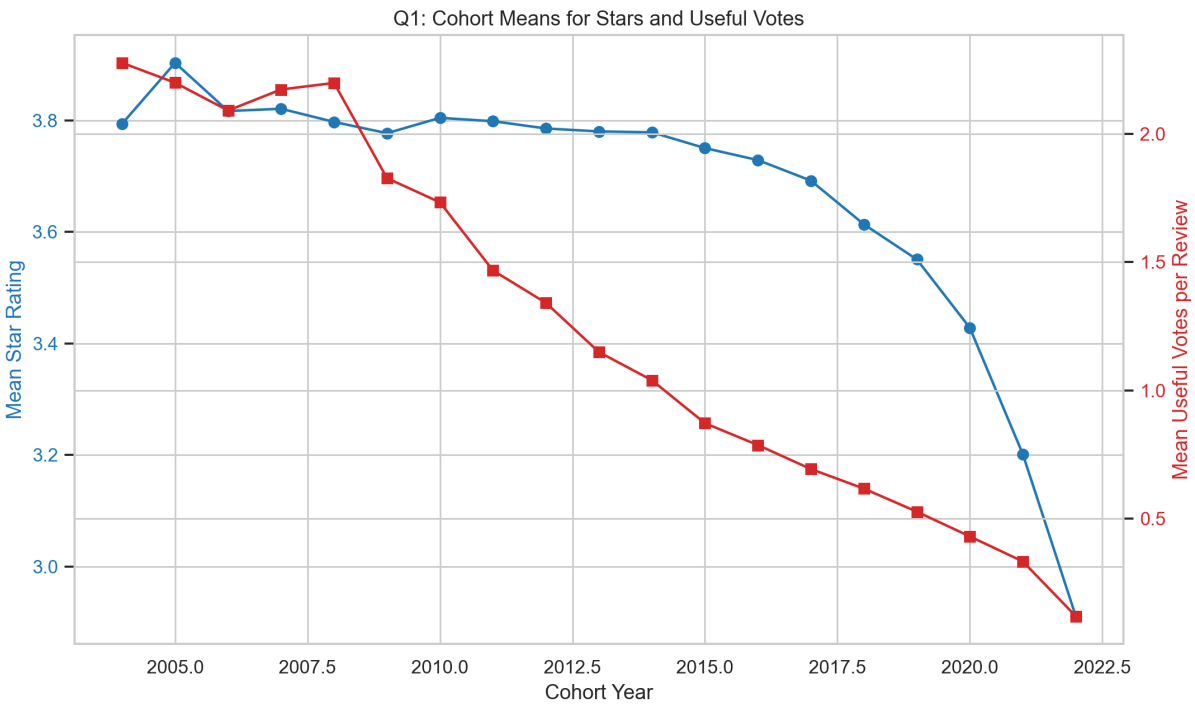

Figure 2: Q1 line plot: mean stars and mean useful votes by cohort year.

Interpretation:

- Older cohorts show stronger useful-vote and rating signals than recent cohorts.
- Later cohorts (notably post-2018) trend lower on both dimensions.

Q2: Month-over-Month Trend Consistency by Category

For each category with at least 500 reviews, monthly average stars are computed and consistency is measured as the proportion of consecutive month-pairs with increase/decrease.

Direction	Category	Consistency Ratio
Upward	Japanese Curry	0.544
Upward	Home Cleaning	0.543
Upward	Beer Gardens	0.541
Downward	Medical Centers	0.566
Downward	Walk-in Clinics	0.554
Downward	Eyelash Service	0.550

Table 2: Q2 most consistent upward and downward category trends.

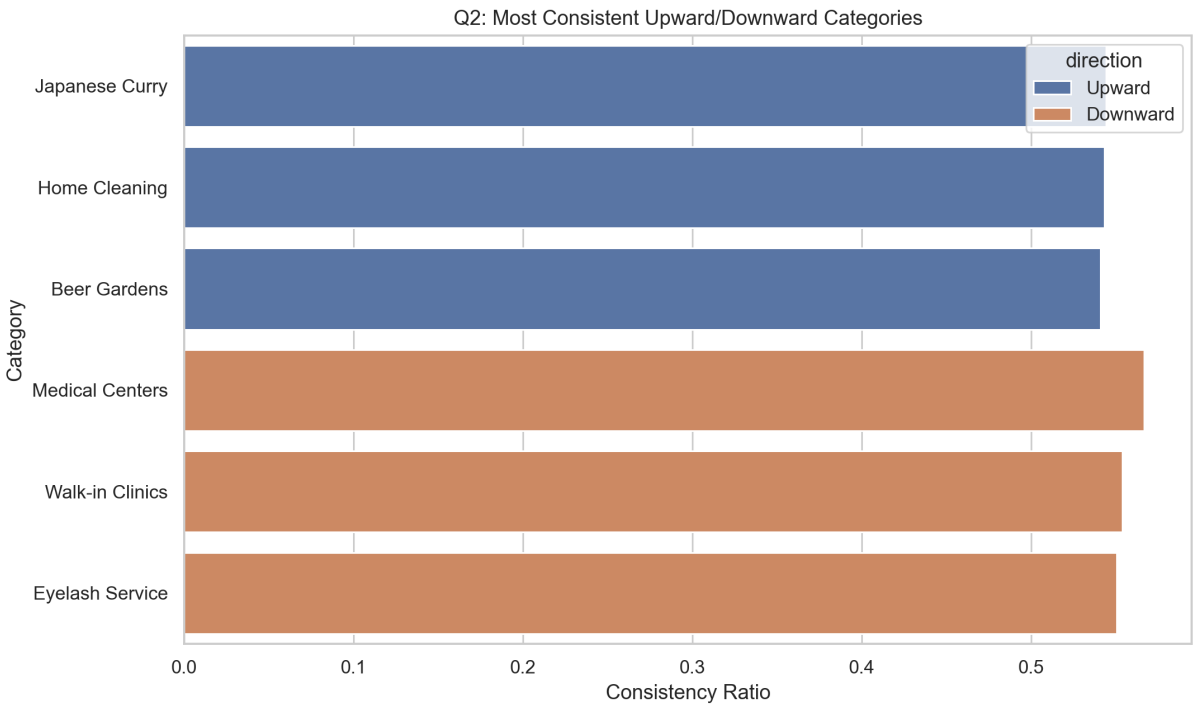

Figure 3: Q2 bar plot: consistency ratios for most upward/downward categories.

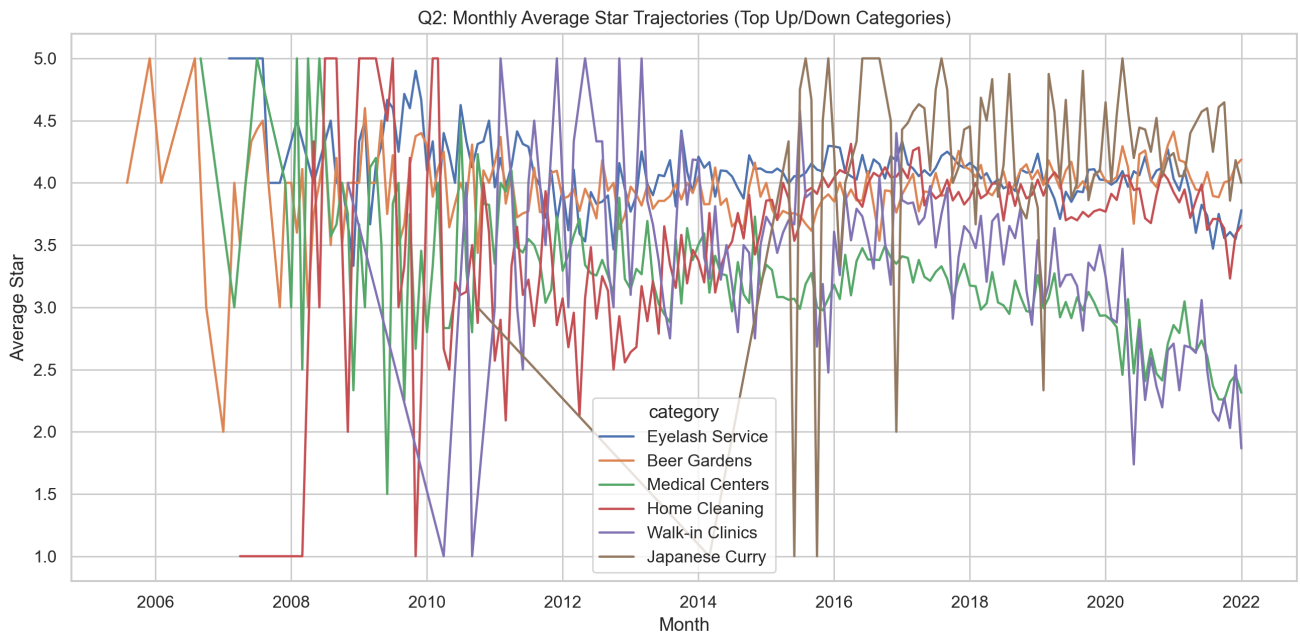

Figure 4: Q2 line plot: monthly trajectories for selected categories.

Interpretation:

- Directional consistency exists, but no category is fully monotonic.
- Both upward and downward groups show meaningful month-to-month volatility.

Q3: Check-in Class Cross-tabulation

Businesses are classified into low/medium/high check-in classes using quartiles from check-in counts (Q1=6, Q3=72). For each top-10 category by total review count, metrics are computed by class: mean stars, mean

review count, and tips-to-reviews ratio.

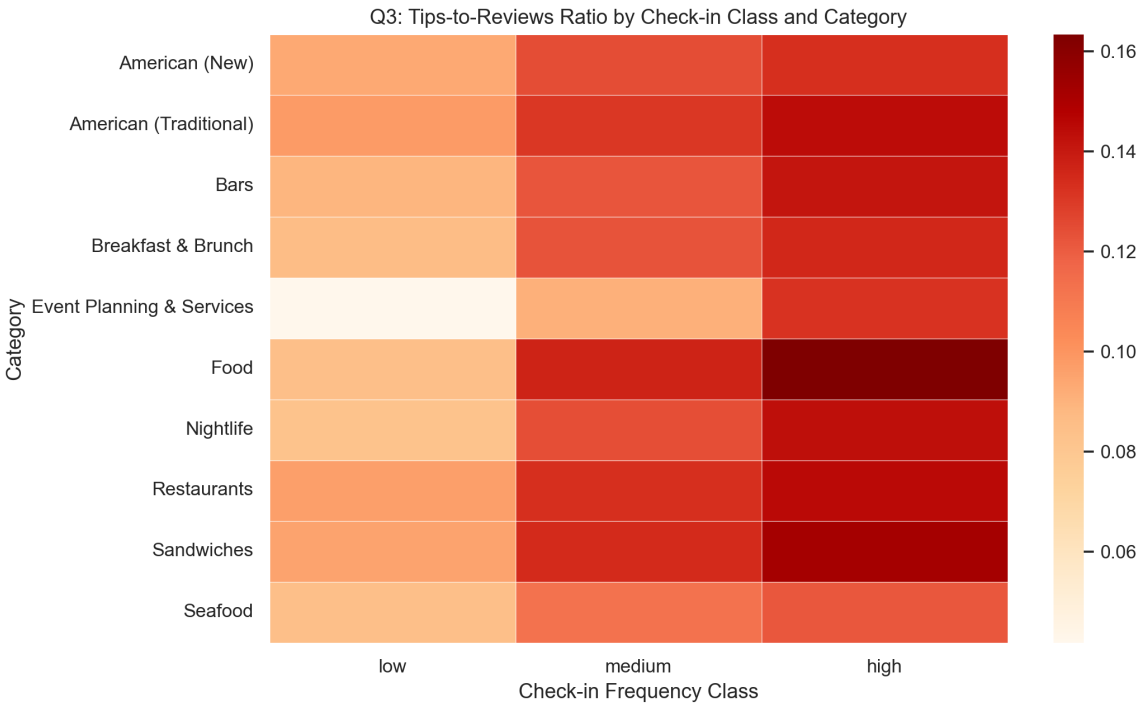

Figure 5: Q3 heatmap: tips-to-reviews ratio by category and check-in class.

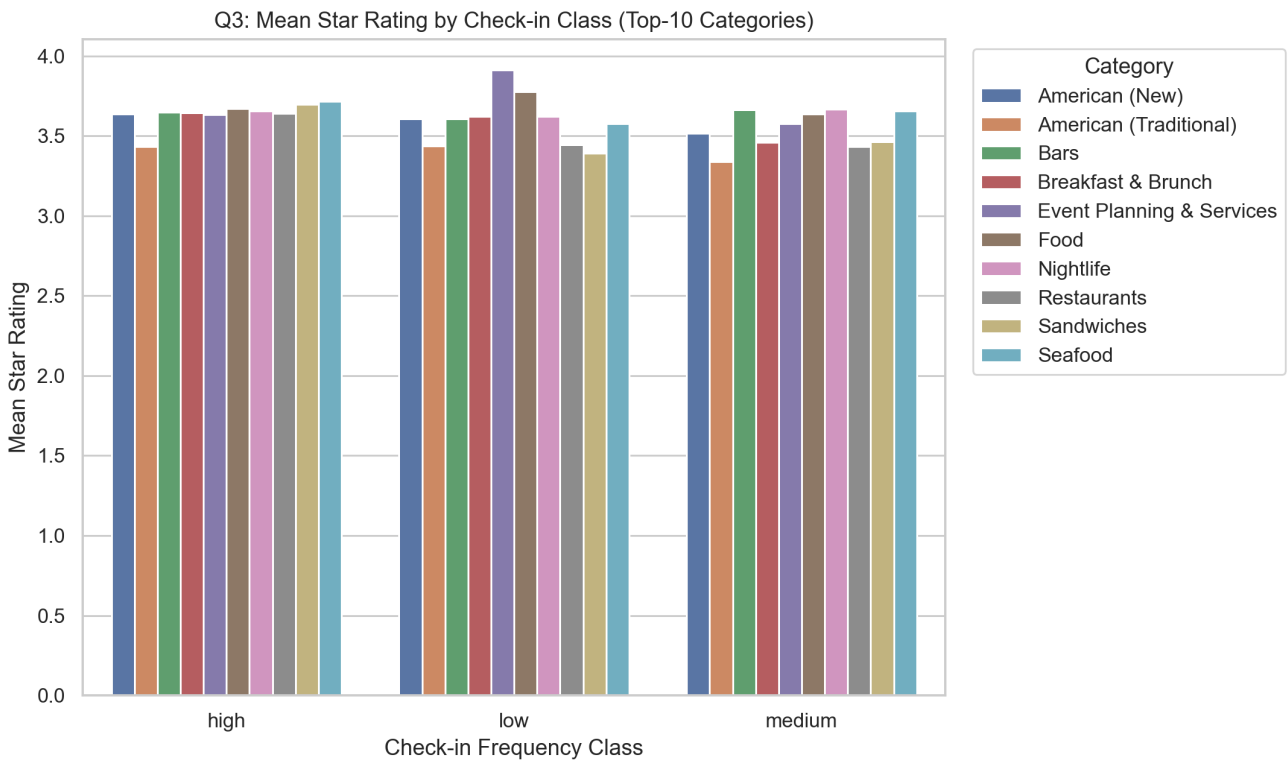

Figure 6: Q3 bar plot: mean stars by check-in class across top categories.

Interpretation:

- High check-in businesses generally have substantially higher mean review volumes.
- Tips-to-reviews intensity is commonly strongest in the high check-in group.

2 Shared Subset Construction (Used for Neo4j and Predictive Modelling)

Why a Shared Subset Was Built

Part 2 includes graph analytics and predictive modelling workloads that are expensive on the full Yelp dump. To keep execution feasible and reproducible on local hardware while preserving assignment complexity, a single materialized subset was built and then reused across both downstream tracks:

- Neo4j + GDS querying
- Predictive modelling dataset construction and evaluation

Using one canonical subset avoids dataset drift between tasks and ensures that model features, graph structures, and exploratory results are all computed from the same entity universe.

Subset Construction Pipeline

I constructed a shared subset with fixed rules and a deterministic sampling process. The workflow was:

1. Scan businesses and keep only valid candidates (non-empty state/city/categories, minimum review count, optional check-in constraint).
2. Allocate a target business budget across top states using proportional quotas with a minimum floor per state.
3. Within each state, select businesses in a round-robin manner across primary categories to improve category diversity.
4. Keep reviews only for selected businesses, with a per-business cap.
5. Keep users referenced by selected reviews.
6. Prune each kept user's friend list to subset users only (keeps social edges coherent inside subset).
7. Keep check-ins and tips for selected businesses.

This produces a smaller but structurally consistent induced multi-collection/multi-graph dataset.

Subset Configuration and Resulting Size

Subset Construction Parameters

Parameter	Value
Target businesses	6000
Max states	12
Min businesses per state	250
Min business review count	15
Max reviews per business	120

Subset Resulting Counts

Entity	Count
Businesses	6,000
Reviews	437,582
Users	289,869
Check-ins	5,634
Tips	147,910

Table 3: Shared subset configuration and resulting size.

State quota allocation preserved geographic spread while still emphasizing high-signal states. Largest state allocations were PA (1250) and FL (1054), with additional representation from TN, IN, MO, LA, AZ, NV, NJ, CA, ID, and AB.

Why This Matters for the Report

All Neo4j and predictive modelling results below were run on this same shared subset (not on the full original dump), so the findings are internally consistent across sections.

3 Neo4j / Cypher Querying

NQ1: PageRank on User-Business Projection

The query projects a directed weighted graph from users to businesses using review stars as weights and runs GDS PageRank with minimum 20 iterations.

Outputs used for analysis:

- Top-15 businesses by PageRank with review count and average stars
- Spearman correlations: PageRank vs review-count ranking; PageRank vs average-star ranking
- Rank-gap analysis for businesses over/under-ranked by PageRank vs simple metrics

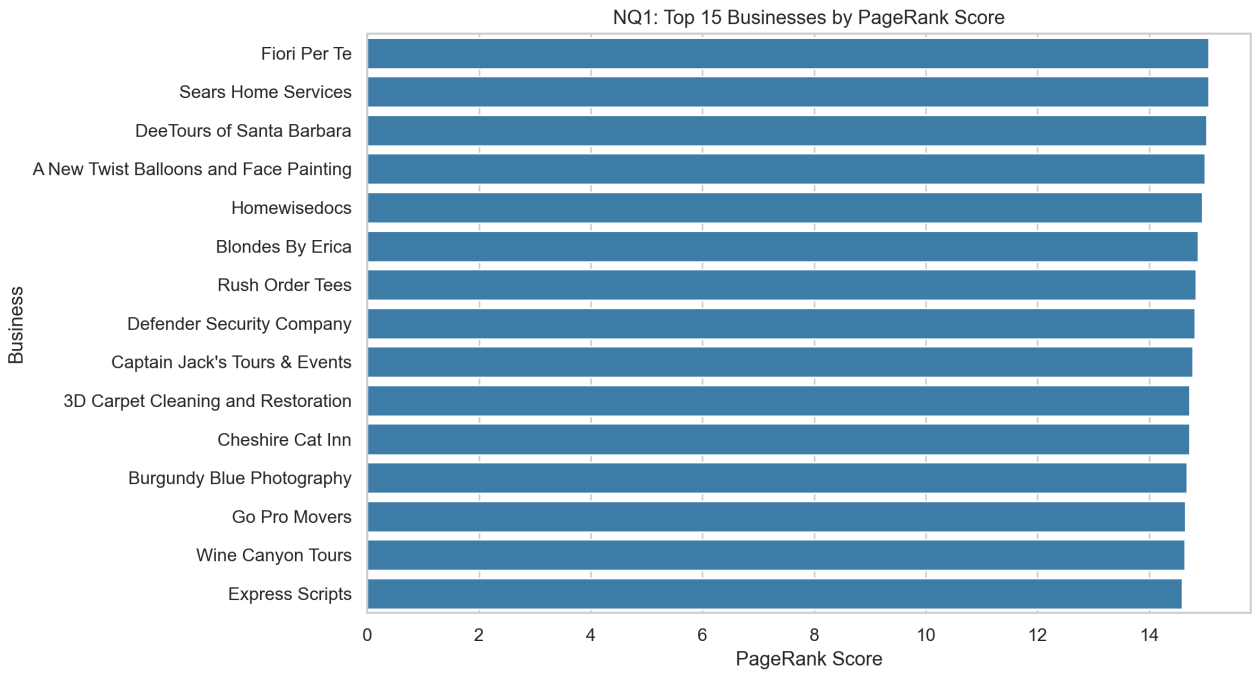

Figure 7: NQ1: Top 15 businesses by PageRank score.

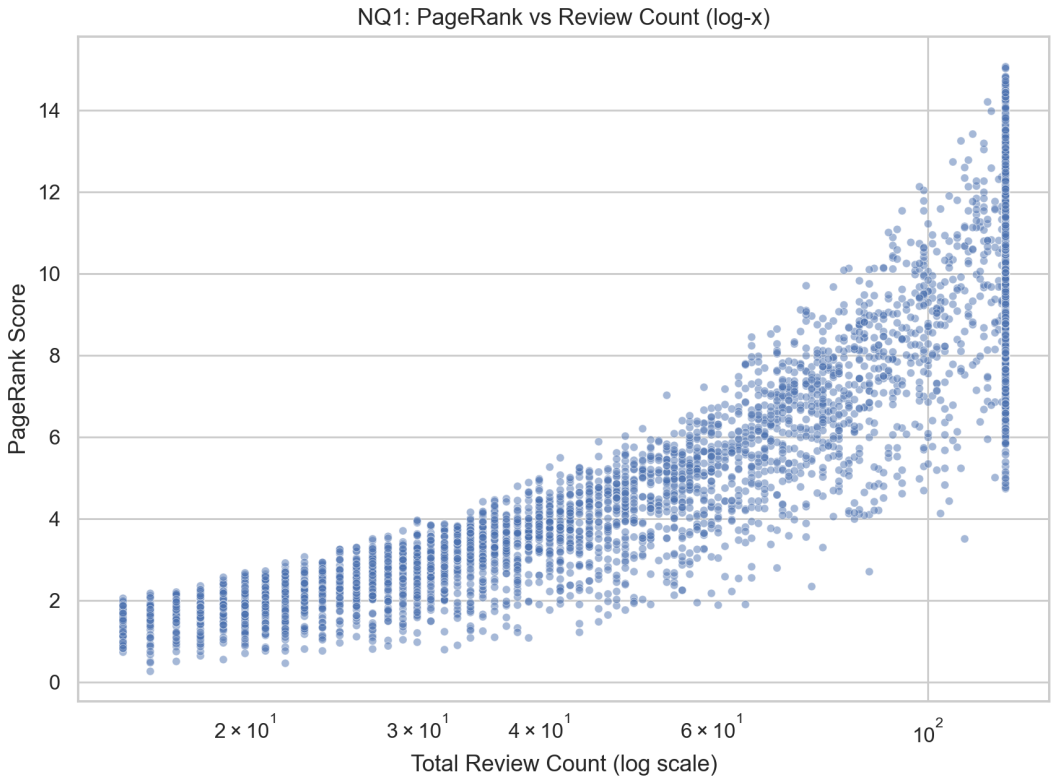

Figure 8: NQ1: PageRank score vs review count.

NQ1 Metric	Value
Spearman(PageRank, review count)	0.929
Spearman(PageRank, average stars)	0.085
Top-ranked business by PageRank	Fiori Per Te

Table 4: NQ1 key findings.

Analytical interpretation:

- The very high PageRank vs review-count correlation indicates that in this subset, **exposure and interaction volume dominate** influence. Businesses with many inbound user interactions absorb centrality mass regardless of sentiment.
- The weak PageRank vs mean-stars correlation shows that **network prominence is not equivalent to customer satisfaction**. A business can be structurally central while being poorly rated, and vice versa.
- Rank-gap diagnostics reveal two distinct regimes:
  - **Over-ranked by PageRank** (large positive gap): entities such as Dent Artist Paintless Dent Removal and Better Business Bureau Serving West Florida have moderate review counts but high graph centrality, suggesting they sit in reviewer pathways that connect active user neighborhoods.
  - **Under-ranked by PageRank** (large negative gap): high-volume brands such as Whole Foods Market and Walmart Supercenter receive many reviews but are less central in reviewer flow, consistent with broad but less tightly connected audience overlap.

Business insights:

- For recommendation/ranking strategy, a two-signal policy is preferable: centrality for discoverability and rating for quality control.
- For business growth, partnerships with highly connected reviewer segments may improve visibility faster than marginal rating improvements alone.

NQ2: Louvain Community Detection

I ran Louvain on the friendship graph, filtered communities with at least 25 members, and profiled each community by top states, top categories, and geographic concentration index (GCI).

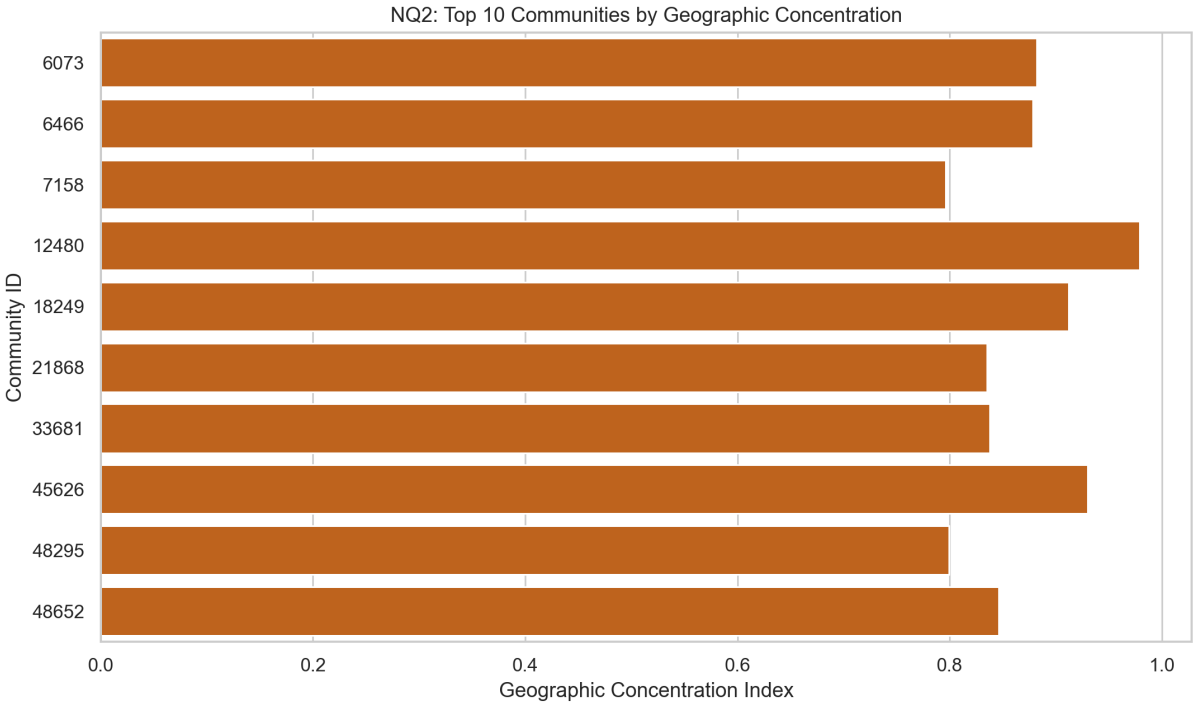

Figure 9: NQ2: Top communities by geographic concentration index.

NQ2 Metric	Value
Communities with size ≥ 25	19
Most concentrated community (GCI)	12480 (0.979)
Least concentrated community (GCI)	7449 (0.171)
Selected projection threshold	Minimum 10 reviews

Table 5: NQ2 key findings.

Analytical interpretation:

- GCI spans a wide range (approximately **0.17 to 0.98**), which means the social graph contains both highly local and highly cross-region communities.
- The most concentrated communities are nearly single-state clusters (for example community 12480 is AB-dominant), whereas the least concentrated communities distribute review activity across multiple states with no dominant geography.
- Across communities, top reviewed categories repeatedly include Restaurants, Food, and Nightlife, implying that **community differentiation is more geographic than categorical**. In other words, members in different communities often consume similar category types but in different places.

Business insights:

- Geo-targeted campaigns should prioritize high-GCI communities because local message relevance is likely to be strongest.
- Cross-state expansion and referral strategies are better aligned to low-GCI communities where social influence is less geographically concentrated.

NQ3: Node Similarity (Jaccard) for Market Saturation

Node Similarity is run on business nodes via shared reviewer neighborhoods. Threshold starts at 5 common reviewers and falls back to 4 or 3 if needed. Results include top-5 most saturated and top-5 most fragmented city-category combinations, excluding combinations with fewer than 5 businesses.

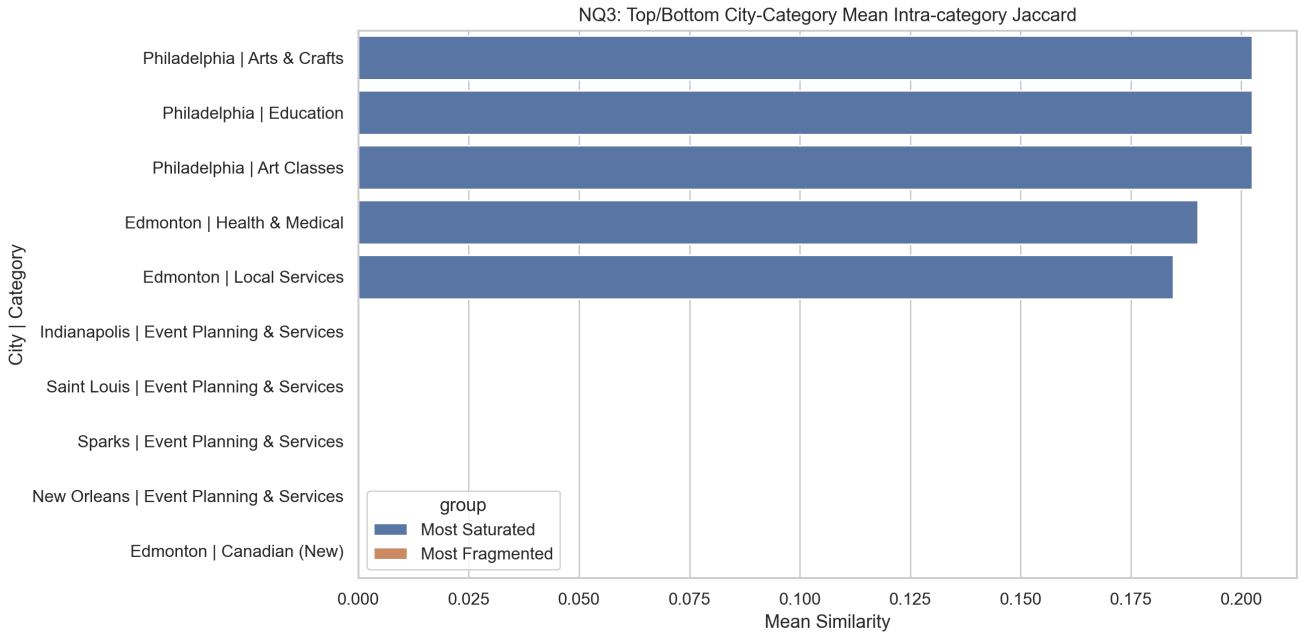

Figure 10: NQ3: Most saturated and most fragmented city-category combinations.

NQ3 Metric	Value
NodeSimilarity setting	top-k 5, min common reviewers 5
Most saturated combo	Philadelphia — Arts & Crafts (0.203)
Most fragmented combo	Indianapolis — Event Planning & Services (0.000)

Table 6: NQ3 key findings.

I also compared mean stars, star spread, and mean review count for the most and least saturated combinations.

Analytical interpretation:

- Saturation (high intra-category audience overlap) appears in narrow pockets. Example: Philadelphia — Arts & Crafts has mean similarity **0.203**.
- Many city-category pairs have similarity near zero, including several Event Planning & Services combinations. This indicates fragmented demand where different businesses draw largely distinct reviewers.
- Comparing the extreme pairs:
  - Saturated pair (Philadelphia — Arts & Crafts): mean stars **4.08**, star standard deviation **0.845**, mean review count **112.9**.
  - Fragmented pair (Indianapolis — Event Planning & Services): mean stars **4.13**, star standard deviation **0.701**, mean review count **188.0**.
- In this sample, the saturated market has **higher rating variance but lower average review volume**. This supports the interpretation that direct audience overlap can intensify competitive differentiation while limiting per-business attention.

Business insights:

- Entering saturated categories requires sharper positioning because consumers compare overlapping options directly.
- Entering fragmented categories may benefit from local niche capture and channel-specific acquisition rather than direct head-to-head brand competition.

NQ4: Betweenness vs Degree Centrality

Betweenness and degree are computed on the social graph. The top-20 overlap is reported, and high-betweenness-low-degree users are compared against high-degree users on review count, distinct cities reviewed, and distinct categories reviewed.

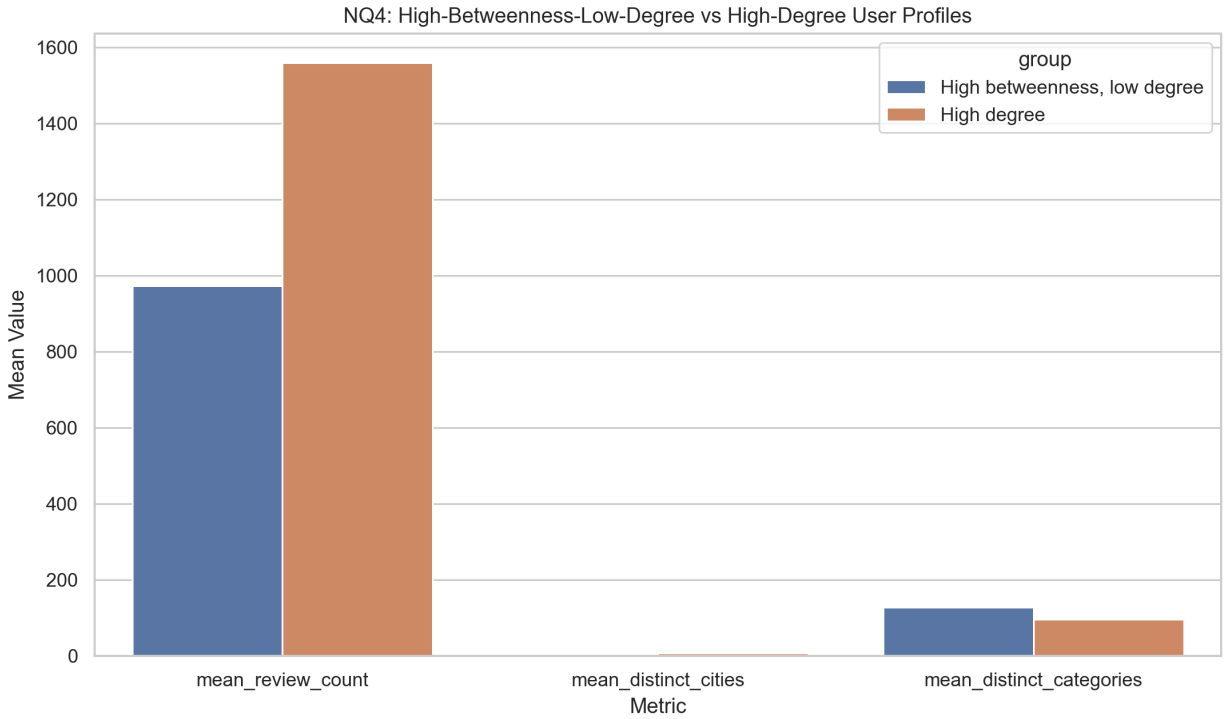

Figure 11: NQ4: Group-level comparison for centrality-defined user sets.

NQ4 Metric	Value
Top-20 overlap (betweenness vs degree)	16 users
High-betweenness but not top-degree users	4 users
Mean review count (bridge vs high-degree)	973.5 vs 1559.25
Mean category breadth (bridge vs high-degree)	128.25 vs 96.2

Table 7: NQ4 key findings.

Analytical interpretation:

- The 16/20 overlap confirms that many hubs are also bridges, but the 4 non-overlapping high-betweenness users are analytically important.
- Those bridge users are **Isabella, Marcia, Aimee, and Ophelia** (present in top betweenness but not top degree). Their profile suggests connector behavior: fewer total links than hub users, but links that cross community boundaries.
- Compared with high-degree users, they have:
  - lower activity volume (mean review count 973.5 vs 1559.25),

  - lower geographic spread (4.75 vs 7.30 cities),
  - higher category breadth (128.25 vs 96.2 categories).

- This pattern indicates **cross-interest bridges**: users who connect topic spaces more than they maximize local neighborhood size.

Business insights:

- Degree alone is insufficient for influence seeding. Campaigns should include betweenness-heavy connectors for cross-community diffusion.
- Category-diverse connectors are valuable for introducing new categories or cross-selling between adjacent verticals.

NQ5: Link Prediction Pipeline for Recommendations

For NQ5, I used a GDS link prediction pipeline over a bipartite User-Business graph where "visit" is modeled by writing a review. The final run follows the required chronological holdout policy: each user's most recent review is marked as test.

NQ5 Metric	Value
AUC-ROC	0.704
Precision@10	0.0035
Split method	Per-user most recent review as test
Fallback thresholds	User 20 reviews, business 10 reviews
Evaluation cap / prediction pool	200 users / topN 6000
Coverage	200 evaluated, 145 with predictions

Table 8: NQ5 evaluation summary.

Five most predictive features used by the final recommendation model:

- HADAMARD interaction over FastRP embeddings
- COSINE interaction over FastRP embeddings
- L2 distance interaction over FastRP embeddings
- FastRP node embedding representation (64 dimensions)
- Logistic regression scoring layer

Sampled User	Top-3 Recommendations
-FZBTk...	Tucson Asian Rejuvenation Massage Spa; Honky Tonk Central; The Tampa Riverwalk
-G7Zk...	Carvana Philadelphia; Clayton's Hunting & Fishing; Frontier
0G-QF...	Le Jouet; Casablanca Restaurant; The Home Depot
0Igx-...	Erzulie's; Road Kill; Trader Joe's
0zDHi...	New Orleans Museum of Art; Taj Mahal Indian Cuisine; Jamie Hayes Gallery

Table 9: Top-3 recommendations for five sampled users (NQ5).

Improvements applied:

- Enforced strict bipartite user-to-business prediction constraints to avoid invalid pair types.
- Switched to an evaluation cohort of users with both train and test edges.
- Increased a bounded prediction pool and then computed per-user top-10 ranking.

- Computed Precision@10 over all evaluation users, assigning zero to users without returned predictions.

Interpretation:

- AUC-ROC is moderate (0.704), showing the model has some discrimination ability on scored candidates.
- Precision@10 remains very low (0.0035), indicating weak per-user recommendation usefulness in the current configuration.
- Only 145 of 200 evaluation users received scored recommendations in the returned pool, which directly depresses Precision@10 under the assignment-consistent averaging rule.
- In the final setup, user-business affinity signals dominate because the model primarily learns from interaction structure and embedding-based similarity in the bipartite graph.
- Social-graph influence is comparatively weak in this NQ5 run because explicit friend-network features are not directly injected into the prediction stage; this is likely one reason why ranking quality at top-10 remains limited despite reasonable global discrimination.

4 Predictive Modelling

Objective and Data Scope

Goal: predict the number of useful votes each review receives.

Data source is the same shared subset used throughout Part 2. The final modeling table contains:

Predictive Data Metric	Value
Rows	437,579 reviews
Columns	30
Source	Shared Yelp subset used in Part 2

Table 10: Predictive modelling data scope.

Subset Distribution Diagnostics

To understand target skew before modeling, I analyzed the distribution diagnostics:

- bucket-level count matrix
- distribution summary statistics

The target is strongly right-skewed:

Target Distribution Summary

Statistic	Value
Mean	1.471
Median	0
P95	6
P99	14
Max	539

Test-set Stratification (Required Buckets)

Bucket	Count	Share
0	45,246	51.70%
1–5	37,045	42.33%
6–20	4,854	5.55%
21+	371	0.42%

Table 11: Useful-vote distribution and stratified test-set composition.

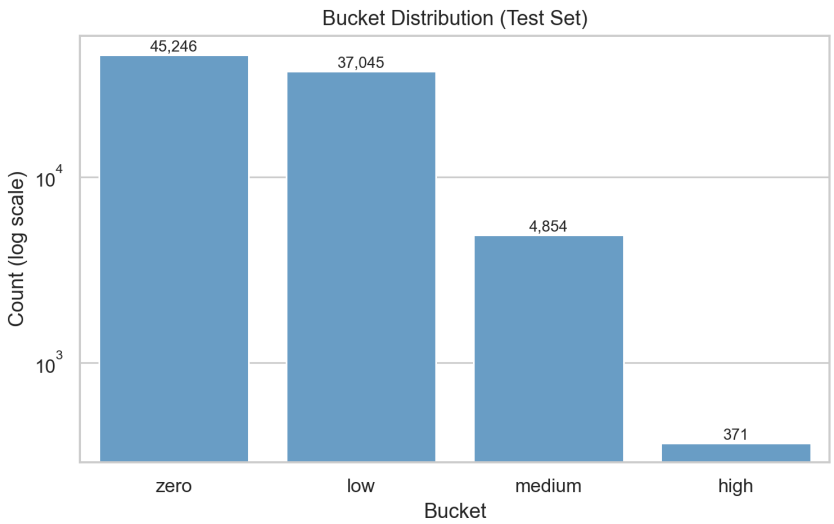

Figure 12: Predictive modelling: test-set bucket distribution (log scale).

Figure 13: Predictive modelling: train vs test target distribution.

Feature Engineering (with Source/Formula)

The assignment requires review-level, user-level, and graph-derived features. The final model matrix uses 28 numeric features (18 base + engineered interactions/transforms). Table below summarizes representative features and construction.

Feature	Source	Definition / Formula
review stars	Mongo reviews	Star rating of the review.
review text length	Mongo reviews	Character length of review text.
review word count	Mongo reviews	Token count in review text.
review recency (days)	Mongo reviews	Days from review date to latest review date in subset.
user review count	Mongo users	Total historical reviews by user profile.
user tenure (days)	Mongo users + review date	Review date minus user yelping-since date.
user elite years count	Mongo users	Number of elite years listed for user.
user friends count	Mongo users	Friend-list length after subset pruning.
business review count	Mongo businesses	Historical business review volume.
business average stars	Mongo businesses	Profile-level mean business stars.
graph user degree	Neo4j graph	User degree in the friendship social graph.
graph user community size	Neo4j NQ2 output	Louvain community size linked via user community id.
graph user community GCI	Neo4j NQ2 output	Geographic concentration index of user community.
graph business city-category similarity	Neo4j NQ3 output	Mean city-category intra-category reviewer similarity (saturation proxy).
user reviews per tenure day	Engineered	user review count divided by (user tenure days + 1).
review star × business average	Engineered	Interaction between review stars and business average stars.
business popularity × saturation	Engineered	business review count multiplied by city-category saturation similarity.

Table 12: Predictive feature examples with data sources and formulas.

Graph-feature rationale:

- Graph user degree: users with denser social neighborhoods plausibly produce reviews with broader exposure.
- Graph user community GCI: community structure and geographic concentration can influence downstream visibility and relevance.
- Graph business city-category similarity: market saturation affects how comparable and discoverable reviews are within category-city niches.

Modelling Strategy and Methods

Train/test split follows the required stratification buckets: {0, 1–5, 6–20, 21+}. Candidate models were evaluated in an auto-selection pipeline:

- Histogram Gradient Boosting regressors (standard and log-target variants)
- Poisson-loss gradient boosting variants
- Tweedie/Poisson GLM baselines
- Fast random forest baseline
- Hurdle model (two-stage: classify useful>0, then regress positive counts)
- Blend candidates combining complementary predictors

Model objective selection used minimum RMSE on the held-out test set. Feature importance is reported via permutation importance (on sampled test rows) for interpretability.

Model Comparison and Final Selection

I compared candidate models using a consolidated metrics matrix and a line-chart visualization of RMSE, MAE, and R^2.

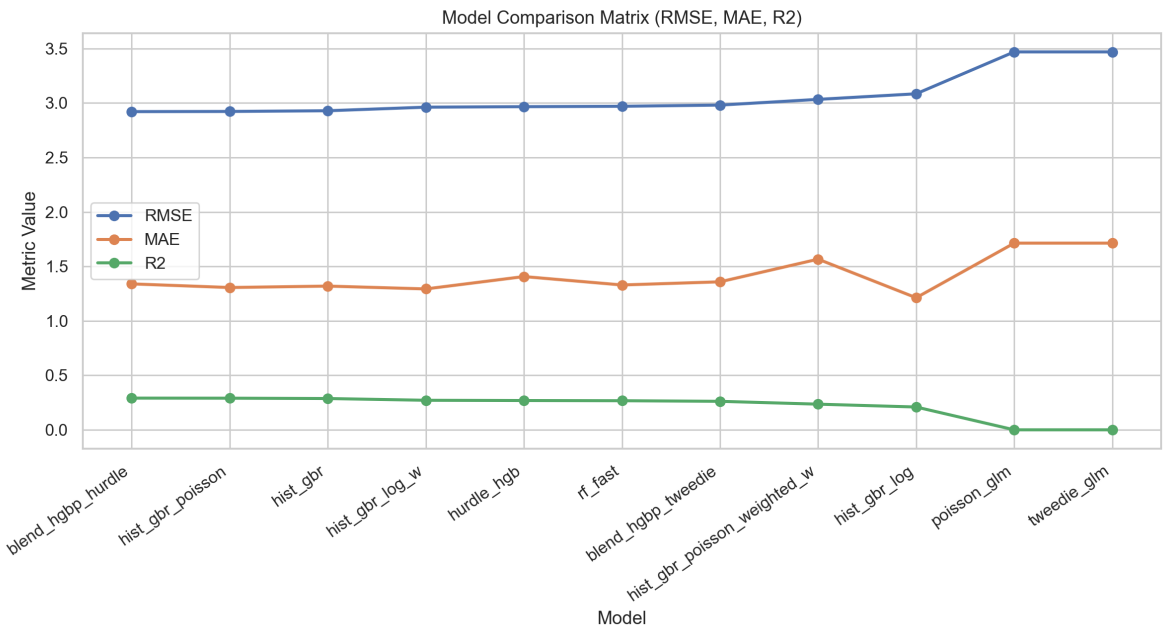

Figure 14: Predictive modelling: RMSE/MAE/R^2 across candidate models.

Best model selected: **blend_hgbp_hurdle**

Metric	Value
RMSE	2.9204
MAE	1.3394
R^2	0.2911

Table 13: Final predictive model performance.

The blend achieved the lowest RMSE among all candidates, slightly improving over the best single gradient-boosting model.

Required Bucket-wise Evaluation

Bucket-wise results (required groups: 0, 1–5, 6+):

Bucket	RMSE	MAE
0 useful votes	1.3312	0.8937
1–5 useful votes	1.7067	1.1706
6+ useful votes	10.3371	6.3953

Table 14: Bucket-wise predictive performance.

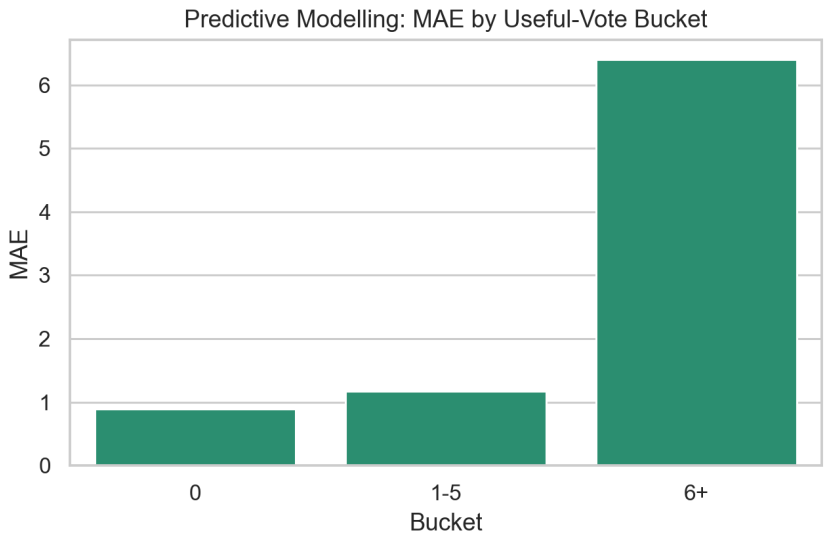

Figure 15: Predictive modelling: MAE by useful-vote bucket.

Interpretation:

- Performance is strongest in the dominant low-count regime (0 and 1–5 buckets).
- Error increases sharply in 6+ due to extreme sparsity and heavy-tail variance, which is expected in useful-vote prediction.

Prediction Diagnostics

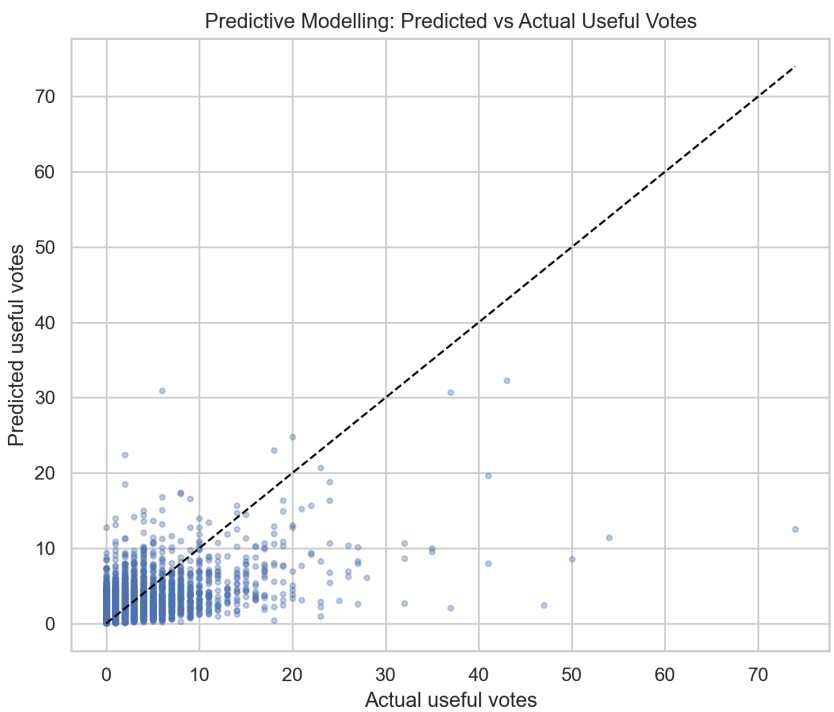

Figure 16: Predictive modelling: predicted vs actual useful votes.

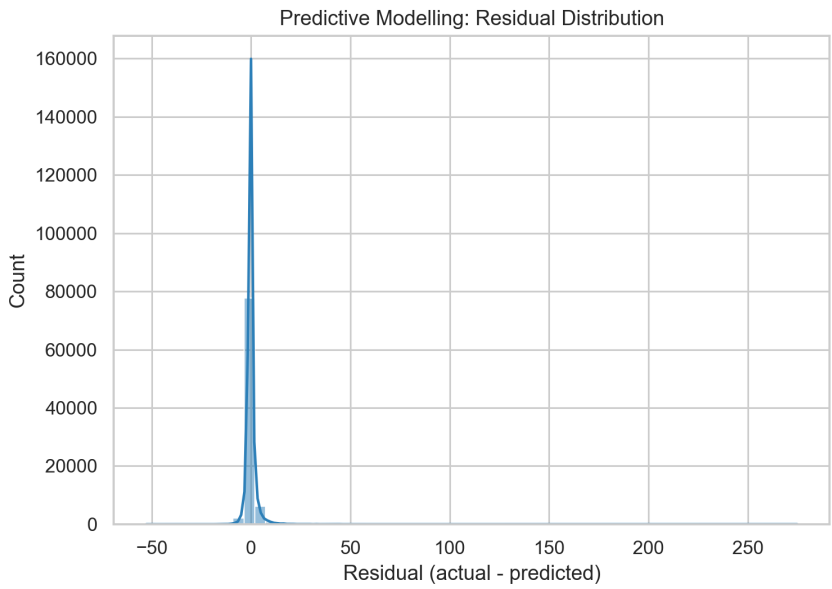

Figure 17: Predictive modelling: residual distribution.

The scatter and residual plots show central-mass fit quality but persistent tail underfit in high useful-vote reviews, consistent with bucket-wise results.

Feature Importance and Discussion

Top-3 predictive features (permutation importance):

Feature	**Importance**
Review recency in days	0.2156
Business review count	0.1198
Review text length	0.1071

Table 15: Top-3 predictive features by permutation importance.

Data-supported interpretation:

- **Recency** dominates, suggesting platform attention dynamics are time-sensitive (newer/temporally positioned reviews attract different engagement levels).
- **Business review volume** indicates exposure effects: reviews for more visible businesses tend to receive more interactions.
- **Text length** captures informational richness; longer reviews often provide more evaluative content and can attract more useful votes.

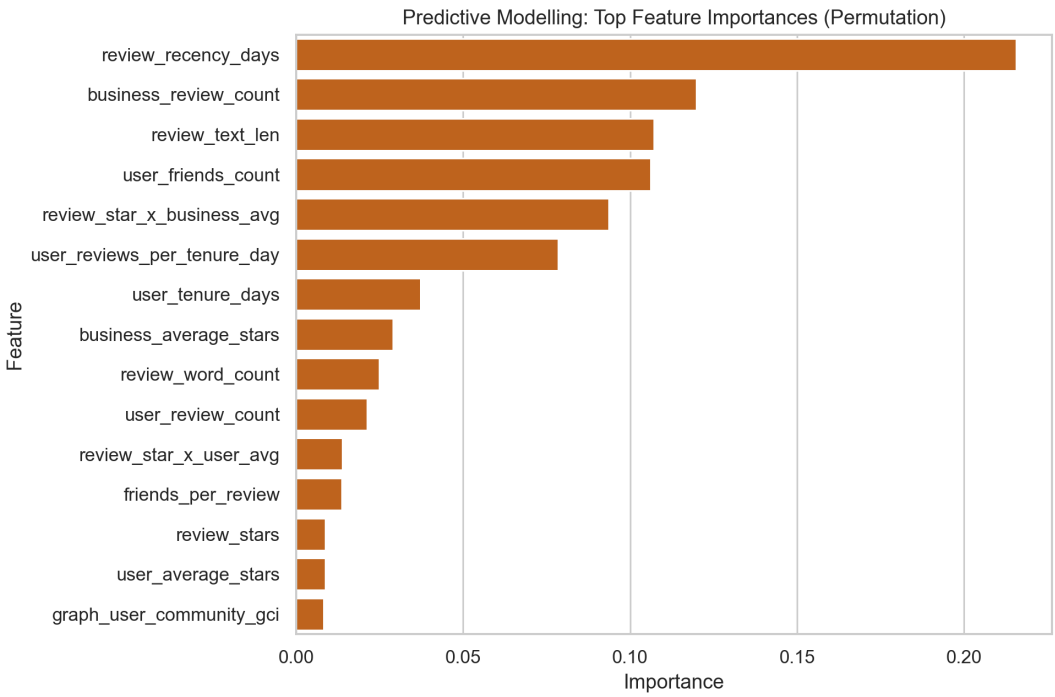

Figure 18: Predictive modelling: top feature importances.

Limitation and Potential Consequence

Limitation: the target is highly imbalanced with very few extreme high-useful instances (21+), and even fewer very large values. This limits tail calibration and reduces reliability for ranking the very top reviews.

Potential consequence if deployed for ranking: an under-calibrated model may systematically under-rank niche but high-value reviews (especially from smaller businesses or less represented communities), reinforcing popularity bias toward already visible entities.

5 Query Text Files

Complete query statements are provided separately in the submission package.